

Active inference and the grounding problem: Toward a more general theory of intelligence

Abstract: Active inference frameworks can be represented in terms of the tuple $(P, Q, R, S, A, U, \Omega)$, where P denotes the generative model of the agent, Q denotes the approximate posterior, and R denotes the generative process. The first part of our argument involves demonstrating that active inference frameworks have to confront a grounding problem: either R is given exogenously (Horn 1 or primitive grounding), R is derived from internal mechanisms (Horn 2 or non-primitive grounding), or R is ultimately reducible to the probabilistic structure continuous with P (Horn 3 or groundlessness). The second part of our argument involves solving this grounding problem through a development of our account of intelligence. In particular, our account of intelligence, nested within a theory of structure and a theory of life, allows us to provide a non-primitive, non-circular, and non-reductive grounding of evaluative structure.

Keywords: Active inference; free energy minimization; Bayesian; trilemma; theory of intelligence

§1 Introduction

Active inference – largely developed by the British neuroscientist Karl Friston and his collaborators – is often widely regarded as a unified theory of intelligence, behaviour, and life. Owing to the extraordinary scope of these frameworks, spanning physics, biology, cognitive science, and even social science, at least some philosophers have suggested that active inference frameworks approach the status of a theory of every thing (Hipólito, 2019). At the core of these frameworks is a free energy minimization principle, which provides a formal imperative designed to capture the dynamics of self-organizing systems operating across multiple scales. According to active inference frameworks, an intelligent system is, roughly, any system that behaves as if she minimizes expected free energy relative to a set of prior preferences.

In this paper, we shall first outline a mathematical formalism, representing active inference in terms of the tuple $(P, Q, R, S, A, U, \Omega)$ and policy selection in terms of the expected free energy function $G(\pi)$ (§§ 2–3). As this mathematical formalism is value-neutral, it cannot by itself determine which states ought to be preferred. The worry that active inference frameworks presuppose their normativity, as opposed to grounding it, is then articulated in the form of a lemma (normative grounding), theorem (grounding problem), and corollary (incompatibility result) in § 4. At length, we shall propose a solution to the grounding problem. This solution, nested within a theory of structure and a theory of life, is essentially a theory of intelligence that extends active inference frameworks and provides a non-stipulative grounding of value (§§ 5–6).

§2 A *précis* of active inference

Active inference frameworks typically assume the existence of a valuation structure, encoded in prior preferences, and model behaviour as the minimization of surprise, free energy, or uncertainty relative to these priors (Adams, Shipp, and Friston, 2013; Carhart-Harris and Friston, 2010; Clark, 2013; Friston, 2005, 2009; Friston and Stephan, 2007; Hohwy, 2013). The agent functions as a Bayesian entity with prior preferences over a set of states, beliefs, an internal model that is formed on the basis of these beliefs, and an ability to update these beliefs in an evidence-sensitive manner.

More precisely, where $\tilde{o} \in \Omega$ denotes a sequence of observations, $\tilde{s} \in S$ denotes a sequence of hidden or external states of the world, $\tilde{a} \in A$ denotes a sequence of actions that the agent can perform in the world, and the trajectory extends to the present moment t , R denotes the generative process up to t , describing how hidden states evolve, how these hidden states generate observations, and how actions intervene on the world:

$$R(\tilde{o}, \tilde{s}, \tilde{a}) = Pr(\{o_1, \dots, o_t\} = \tilde{o}, \{s_0, \dots, s_t\} = \tilde{s}, \{a_1, \dots, a_{t-1}\} = \tilde{a})$$

R is essentially the true probability distribution over past and present sequences up to t . By contrast, where $u \in U$ denotes a sequence of control states, P denotes the generative model, and the time horizon (typically future-oriented) extends into some future endpoint T , P can be defined in the following manner:

$$P(\tilde{o}, \tilde{s}, \tilde{u}|m) = Pr(\{o_1, \dots, o_T\} = \tilde{o}, \{s_0, \dots, s_T\} = \tilde{s}, \{u_t, \dots, u_T\} = \tilde{u}), \text{ with parameters } \theta \text{ of the model } m$$

P is essentially the agent’s internal probabilistic model of how these observations arise. Furthermore, the shift from t (the present moment) to T (how far into the future the agent is modelling) marks a transition from describing, relative to the generative process R , reality up to t to simulating, relative to the generative model P , the internal model m , and m ’s parameters θ , possibilities up to T . Finally, a policy $\pi \in \{1, \dots, K\}$ refers to an index for a sequence of future control states, such that $(\tilde{u}|\pi) = (u_t, \dots, u_T|\pi)$. Each policy – for instance, π_1 for ‘turn left’, π_2 for ‘wait’, π_3 for ‘turn right’, and so on – represents a possible plan for the future. Where $\hat{s}$ denotes the expected hidden states and $\hat{\pi}$ denotes the expected policy, the approximate posterior Q over hidden states and control states can be defined in the following manner:

$$Q(\tilde{s}, \tilde{u}) = Pr(\{s_0, \dots, s_T\} = \tilde{s}, \{u_t, \dots, u_T\} = \tilde{u}), \text{ with parameters } (\hat{s}, \hat{\pi})$$

Q is essentially the agent’s beliefs over hidden states and future actions, with future actions being structured by a policy π that selects a sequence of control states. The world evolves in accordance with the generative process R (reality), the Bayesian agent situated in the world is equipped with a generative model P (model of reality), this agent observes data $\tilde{o}$, she updates her beliefs in accordance with the approximate posterior Q (beliefs about reality), evaluates candidate policies π in accordance with Q , and selects actions that minimize free energy. All things considered, active inference can be characterized in terms of the tuple $(P, Q, R, S, A, U, \Omega)$ (Friston, Rigoli, et al., 2015).

§3 Divergence and policy selection

The approximate posterior Q is an online approximation, computed within the Bayesian agent at each timestep. Q represents the agent’s beliefs over hidden states and future actions, updated to minimize variational free energy. In addition, Kullback-Leibler divergence D_{KL} is a measure that allows us to quantify the statistical distance between two probability distributions (Kullback, 1997; Kullback and Leibler, 1951).¹ The Bayesian agent aims to minimize the divergence between its approximate posterior Q and the generative model P :

$$D_{\text{KL}}\left(Q(\tilde{s}, \tilde{u}) \parallel P(\tilde{o}, \tilde{s}, \tilde{u} \mid m)\right) = \sum_{\substack{\tilde{s} \in S \\ \tilde{u} \in U}} Q(\tilde{s}, \tilde{u}) \log \frac{Q(\tilde{s}, \tilde{u})}{P(\tilde{o}, \tilde{s}, \tilde{u} \mid m)}$$

This is also known as variational free energy F :

$$F = D_{\text{KL}}\left(Q(\tilde{s}, \tilde{u}) \parallel P(\tilde{s}, \tilde{u} \mid \tilde{o}, m)\right) - \ln P(\tilde{o} \mid m) \geq -\ln P(\tilde{o} \mid m)$$

The agent maintains its approximate posterior Q over hidden states $s \in S$ and updates Q at each timestep by minimizing variational free energy F relative to its generative model P . This Bayesian updating ensures that beliefs remain internally consistent and aligned with observed outcomes $o \in \Omega$ under some chosen policy or sequence of control states π :

$$Q(\tilde{s}_t \mid \tilde{o}_{1:t}, \pi) = \arg \min_Q F(Q \parallel P)$$

¹ A lower D_{KL} value indicates a closer alignment between model and reality (lower surprise or likelihood of prediction error), whereas a higher D_{KL} value reflects a greater mismatch between model and reality (higher surprise or likelihood of prediction error).

Two divergences matter when evaluating how good a policy is:

(Pragmatic divergence) How much divergence is there between the inferred state and the target or preferred state, when averaging over all possible outcomes of policy π ?

(Epistemic divergence) How much divergence is there between the predicted outcomes of a policy π and those outcomes that are plausible, coherent, or well-explained relative to her model P ?

These two divergences (pragmatic and epistemic) give rise to two distinct measures: risk and ambiguity. Risk is a measure of the degree of pragmatic divergence, whereas ambiguity is a measure of the degree of epistemic divergence. These two measures can be formally represented in the following manner:

$$\begin{aligned} \text{(Risk)} \quad & \mathbb{E}_{Q(\tilde{o}|\pi)} \left[D_{\text{KL}}(Q(\tilde{s} | \tilde{o}, \pi) \| R(\tilde{s} | \tilde{o})) \right] \\ \text{(Ambiguity)} \quad & D_{\text{KL}}(Q(\tilde{o} | \pi) \| \tilde{P}(\tilde{o} | \pi)) \end{aligned}$$

Let $G(\pi)$ denote the expected free energy of policy π or – in slightly plainer terms – how good this policy or future plan is, relative to the agent’s model P . We can define $G(\pi)$ as a sum of these two measures:

$$G(\pi) = \underbrace{\mathbb{E}_{Q(\tilde{o}|\pi)} \left[D_{\text{KL}}(Q(\tilde{s} | \tilde{o}, \pi) \| R(\tilde{s} | \tilde{o})) \right]}_{\text{risk}} + \underbrace{D_{\text{KL}}(Q(\tilde{o} | \pi) \| \tilde{P}(\tilde{o} | \pi))}_{\text{ambiguity}}$$

A policy is therefore good, just in case it minimizes $G(\pi)$:

² A low $G(\pi)$ value is associated with a good policy, whereas a high $G(\pi)$ value is associated with a bad policy.

$$\pi^* = \arg \min_{\pi} G(\pi)$$

A good policy (π^*) is low-risk and unambiguous. A policy (good or bad) is low-risk, just in case:

$$\begin{aligned} Q(\tilde{s} \mid \tilde{o}, \pi) &\approx R(\tilde{s} \mid \tilde{o}) \\ Q(\tilde{o} \mid \pi) &\approx \tilde{P}(\tilde{o} \mid \pi) \end{aligned}$$

In other words, a policy is good, just in case this policy produces states that are close to the target or preferred state described by R (pragmatic value) and the predicted observations are reliable (epistemic value).

§4 The grounding problem in active inference

As discussed in § 2, active inference frameworks generally distinguish between P (a generative model), Q (an approximate posterior), and R (a generative process). R governs the evolution of states and observations in the world, P encodes an agent’s internal probabilistic representation of the process R , and Q captures the agent’s own evolving beliefs. Since active inference is defined through the minimization of variational free energy, a distinction is presupposed between the way that the world is (captured by R) and how this world might be modelled by a Bayesian agent (captured by P and Q). Whereas the internal roles of P and Q are relatively well-defined, the theoretical status of R remains underdetermined. This is problematic, especially since R appears to carry an evaluative or preference-like structure that could help us to make sense of pragmatic (non-epistemic) value in the context of $G(\pi)$. More generally, these considerations suggest a minimal requirement for theories of agency:

(Lemma or normative grounding) Any theory of agency that purports to explain goal-directed behaviour must account for the origin of its evaluative structure.

This lemma can be justified through the observation that goal-directed behaviour involves evaluating possible outcomes, distinguishing between better and worse policies, and selecting actions that give rise to different trajectories (§ 3). Therefore, any theory of agency worth its salt must either explicitly treat this evaluative structure as primitive or ground it internally. Our lemma sets the scene for the following theorem:

(Theorem concerning the grounding problem) Within active inference frameworks, the generative process R must satisfy exactly one of the following conditions:

(Horn 1 or primitive grounding) R is given exogeneously, with its evaluative structure stipulated;

(Horn 2 or non-primitive grounding) R is derived from the internal mechanisms of active inference frameworks;

(Horn 3 or groundlessness) R is not distinct from the generative model P , and is ultimately reducible to the probabilistic structure continuous with P .

A structured sketch of a proof of this theorem concerning the grounding problem begins with a survey of the possible grounding relations between R and the resources of active inference frameworks. Since active inference frameworks typically specify a class of generative models P , a principle of free energy minimization governing Q , and an optimization objection defined relative to R , any account of R must either locate its origin outside the frameworks (Horn 1) or within the internal mechanisms (Horn 2) or collapse

it within the probabilistic structure of these frameworks (Horn 3). No fourth option is available, unless these frameworks are extended (see § 5).

Relative to Horn 1, R might be conceded as a brute fact, a hyperprior, a phylogenetic gift, and so on. Given its primitive grounding outside the frameworks, R must then be treated as an input to these frameworks. If evaluative structure is located in R , then this structure becomes a mere matter of stipulation, inherited by P . Normativity gets shifted upstream, active inference frameworks can explain optimization given R but not R itself, and these frameworks are reduced to non-foundational theories of instrumental rationality rather than foundational theories of agency (in accordance with our normative grounding lemma). Certain remarks suggest that at least some active inference frameworks are located at this horn of the trilemma:

[S]election explains *how* biological systems arise; the only outstanding issue is *what* characteristics they must possess. (Friston and Stephan, 2007, p. 423)

Once living organisms are formalized as Bayesian agents minimizing variational free energy, active inference can explain what characteristics these systems possess: epistemic sampling, action-perception loops, self-maintenance in low-surprise states, autopoiesis, and so on. However, active inference frameworks rarely specify, beyond a hand-wavy appeal to evolutionary and developmental history, how these systems arise in general and the characteristics that they must possess to survive or function in particular. This implies that active inference frameworks located at Horn 1 bottom out in something that the frameworks themselves cannot explain.

Relative to Horn 2, suppose that R is non-primitively grounded in or derived from the internal mechanisms of active inference frameworks. This

move, however, invites a form of normative regress: since these internal mechanisms, including free energy minimization F and the generative model P , are defined relative to R , we cannot hope to derive R using F and P without formal circularity.³ Last but not least and relative to Horn 3, suppose that we maintain that R is not distinct from P :

$$R(.) \equiv P(.)$$

The rhetoric of defenders of active inference frameworks betrays an oscillation between Horns 1 and 3. In certain instances, phylogeny and ontogeny are invoked when these individuals are pressed on the source of normativity within active inference frameworks (Horn 1). In other instances, preferences are collapsed into expectations when these individuals are pressed on the nature of the difference between R (the ontic) and P (the epistemic) (Horn 3). When at least some attempt is made to move beyond a hand-wavy, stipulative appeal to evolutionary history, active inference frameworks typically hold that the brain mechanisms for supporting the generative model P are themselves a product of evolutionary history:

Rather, our complex predictive abilities (e.g. planning and imagination) emerged gradually (e.g. via phyletic gradualism or punctuated equilibrium) from simpler predictive and error correction loops (e.g. motor and autonomic reflexes) that were already part of the brains of our earlier evolutionary ancestors – and were key to solving adaptive regulation problems ... (Pezzulo, Parr, and Friston, 2022, p. 1)

³ This chicken-and-egg problem cannot be resolved through an appeal to hierarchical models and empirical priors, since these models and priors operate within a valuation structure that they do not constitute.

This appeal to evolutionary history then becomes an account of learning through hierarchical generative models:

... [O]ne criticism of Bayesian treatments is that they ignore the question of how prior beliefs, which are necessary for inference, are formed. However, this criticism dissolves with hierarchical generative models, in which the priors themselves are optimized. In hierarchical models, causes in one level generate subordinate causes in a lower level; sensory data per se are generated at the lowest level [...] Minimizing the free energy effectively optimizes empirical priors (that is, the probability of causes at one level, given those in the level above). Crucially, because empirical priors are linked hierarchically, they are informed by sensory data, enabling the brain to optimize its prior expectations online. (Friston, 2010, p. 129)

However, empirical priors and prior preferences are not the same kind of thing: empirical priors are beliefs about the causal structure of the world in general and the causes of our sensory input in particular. By contrast, prior preferences are evaluative: they encode the space of valued states and are associated with a valuation structure. Our distinction between empirical priors (causal structure) and prior preferences (valuation structure) maps roughly onto the classical distinction between beliefs and desires – or the Humean distinction between the ‘is’ and the ‘ought’ – in metaethics and decision theory. No amount of active inference concerning what the world is like under R can, on its own, generate a specification of what states $s \in S$ are worth attaining. Relative to Horn 3, preferences (target states under R) collapse into predictions (predicted or expected states under P), evaluative distinctions collapse into statistical regularities, goal-directedness gets reduced to

predictive accuracy, and the belief-desire distinction gets eliminated at the limit. Active inference frameworks located at Horn 3 are compelled to surrender non-trivial accounts of both normativity and agency. At the limit, it has even been urged that values themselves disappear when re-described as prior expectations over sensory trajectories:

[...] the quantities optimized by action under value-learning are exactly the same as those optimized by active sampling under the free-energy principle. This means the notion of value per se is redundant and that much of reinforcement and procedural learning can be recast in terms of active inference. Recent developments provide compelling proof-of-principle that active sampling can be used to solve quite complicated problems in optimum control, without the use of value-learning [...] The basic idea is to replace value-functions with prior expectations about sensory trajectories. Action then ensures prior expectations are met and desired states are frequented. (Friston, 2009, p. 299)

Horns 1–3 are jointly exhaustive, since any attempt to ground R must either succeed or fail. If the attempt succeeds, then R can either be primitively grounded outside active inference frameworks (Horn 1) or non-primitively grounded and derived from the internal mechanisms and resources of these frameworks (Horn 2). If the attempt fails, then R is ultimately groundless and reducible to the probabilistic structure continuous with P (Horn 3). Furthermore, as Horns 1–3 are mutually exclusive, it is inevitable that active inference frameworks get pierced by one of these three horns. This gives rise to a corollary to the theorem concerning the grounding problem:

(Corollary or incompatibility result) Active inference frameworks

cannot simultaneously satisfy the following conditions:

Foundational status (violated under Horn 1);

Absence of circularity or normative regress (violated under Horn 2);

A genuine distinction between preference and prediction (violated under Horn 3).

Our lemma (normative grounding), theorem (the grounding problem), and corollary (incompatibility result) do not show that active inference frameworks fail as frameworks. Rather, they demonstrate that active inference presupposes a solution to the grounding problem that it unfortunately does not provide. This, in turn, motivates our search for a grounding of the evaluative structure that is non-primitive, non-circular, and non-reductive.

§5 Structure, life, and intelligence

We contend that our theory of intelligence, interpreted as an extension of active inference frameworks, solves the grounding problem (§ 4). Our theory of intelligence, in turn, is nested within a theory of structure and a theory of life, both of which we have defended elsewhere. According to our theory of structure, the universe $\mathcal{U}$ has an *ante rem* structure that is mathematically possible, physically possible, and actually realized:

$\mathcal{U} \subset \mathcal{P} \subset \mathcal{M}$, where $\mathcal{M}$ denotes the space of all mathematically possible structures and $\mathcal{P}$ denotes the subset of physically possible structures

At the same time, our universe $\mathcal{U}$ can be modelled as a state space:

$\mathcal{U} \equiv (S, \mathcal{F}, \mathcal{G})$, where S denotes the state space, $\mathcal{F}$ denotes the evolutionary operator (telling us how a state evolves over time), and $\mathcal{G}$ denotes the set of all the symmetries of the laws of nature (transformations such as rotations, translations, and so on that leave the laws invariant)

When subsystems of the universe $\mathcal{U}$, governed by symmetric laws constrained by $\mathcal{G}$, are subject to dynamical evolution $\mathcal{F}$ – whether driven toward thermodynamic equilibrium or sustained far from it by flows of energy and matter – instabilities can result in these subsystems settling into organized states whose stabilizer subgroup H is smaller than the full symmetry group $\mathcal{G}$.⁴ Where Φ denotes the gradients of energy, matter, and information driving dissipative structures and H denotes the flux-dependent stabilizer subgroup:

$$H(s, \Phi) \equiv \{g \in \mathcal{G} \mid g \cdot s = s \text{ under dynamics } \mathcal{F}(s; \Phi)\}$$

According to our theory of life, a living system $\mathcal{L}$ in the universe $\mathcal{U} \equiv (S, \mathcal{F}, \mathcal{G})$ exists iff there exists a domain $S_{\mathcal{L}} \subset S$ and a set of processes $\mathcal{F}_{\mathcal{L}} \subset \mathcal{F}$ such that the following axioms hold:

C1 (Self-maintaining state domain): The processes $\mathcal{F}_{\mathcal{L}}$ primarily evolve states within the domain $S_{\mathcal{L}}$: $\mathcal{F}_{\mathcal{L}}(S_{\mathcal{L}}) \subseteq S_{\mathcal{L}}$;

C2 (Organizational closure): There exists a mapping such that every process f has its material substrate or catalyst generated by some state s in $S_{\mathcal{L}}$: $\forall f \in \mathcal{F}_{\mathcal{L}}, \exists s \in S_{\mathcal{L}} : \text{Inst}(s, f)$;

C3 (Autocatalytic closure): There exists a subset $A \subset S_{\mathcal{L}} \times \mathcal{F}_{\mathcal{L}}$,

⁴ At or near the equilibrium (the limiting case), $\mathcal{F}$ selects symmetry-breaking states via potential minimization. Far from the equilibrium (the richer, Prigoginean case), $\mathcal{F}$ selects and maintains symmetry-breaking states via dissipative attractors.

forming an autocatalytic network;

C4 (Thermodynamic coupling): Where the boundary $\partial S_{\mathcal{L}}$ separates internal states ($S_{\mathcal{L}}$) from external states ($S \setminus S_{\mathcal{L}}$), the living system $\mathcal{L}$ maintains local order by exporting entropy faster than it imports it across this boundary $\partial S_{\mathcal{L}}$: $\Phi_{\text{export}} > \Phi_{\text{import}}$ (non-equilibrium thermodynamics). Additionally, where there are multiple gradients – as opposed to the single gradients that sustain non-living dissipative structures – a coupling operator can convert coupled gradients into work: $\exists \Phi_1 \Phi_2 (C : \Phi_1 \otimes \Phi_2 \rightarrow W_{\mathcal{L}})$. This work $W_{\mathcal{L}}$ helps to sustain both $\mathcal{F}_{\mathcal{L}}$ and $S_{\mathcal{L}}$;

C5 (Heritable variation): There exists a compression $S_{\mathcal{L}} \rightarrow G_{\mathcal{L}}$ (a genetic subspace) such that $G_{\mathcal{L}}$ encodes a subset of $\mathcal{F}_{\mathcal{L}}$, $G_{\mathcal{L}}$ is copied with high though imperfect fidelity under $\mathcal{F}_{\mathcal{L}}$, and variation in $G_{\mathcal{L}}$ produces heritable variation in $\mathcal{F}_{\mathcal{L}}$.

§6 From viability to valuation

Active inference frameworks assume a valuation structure but fail to explain why any system should have preferences at all. It follows from our normative grounding lemma (§ 4) that these frameworks therefore fail as a theory of agency. To address this grounding problem, we need a non-stipulative grounding of value. Instead of starting with value *sensu strictissimo*, we propose something weaker and more straightforward:

(Viability) Either a living system $\mathcal{L}$ continues to persist as the kind of system that it is, or it does not.

The persistence of living systems such as $\mathcal{L}$ is not determined by their internal states alone, but also by how they are situated in and coupled to

the world. Where R specifies how hidden states of the world evolve and give rise to observations, viability must be defined relative to R : a state $s \in S_{\mathcal{L}}$ is viable, just in case it can be maintained dynamically by $\mathcal{L}$ under the dynamics induced by R . Relative to our theory of life (§ 5), C1–C4 essentially constitute viability conditions, or conditions for living organisms $\mathcal{L}$ to maintain themselves far from equilibrium and in terms of thermodynamically grounded, dissipative structures. C5, in turn, ensures a selection of these viability conditions across generations. The coupling operation $\exists \Phi_1 \Phi_2 (C : \Phi_1 \otimes \Phi_2 \rightarrow W_{\mathcal{L}})$ in C4 identifies how coupled gradients enable the construction of life-essential constraints and not merely the dissipation of energy. We can now assert that a state $s \in S_{\mathcal{L}}$ is viable, just in case it sustains a gradient-coupling regime in a manner that makes C1–C4 satisfiable.⁵ More generally, let $\mathcal{K} \subset S_{\mathcal{L}}$ refer to the set of all states from which $\mathcal{L}$ can return to its dissipative, gradient-coupling regime. In each of these states, $\mathcal{L}$ persists as the kind of organized entity that it is. For convenience, we can term $\mathcal{K}$ the viability set.⁶

Next, we must understand why viability matters in the context of life. Living systems are dynamic rather than static, and must actively remain within $\mathcal{K}$ even in the face of constant perturbation. To regulate its own behaviour under perturbation, $\mathcal{L}$ must be able to distinguish between better and worse states relative to staying within $\mathcal{K}$. Making these distinctions, in turn, appears to require a valuation structure that allows us to evaluate states $s \in S_{\mathcal{L}}$. Any dynamical, living system $\mathcal{L}$ that reliably returns to a

⁵ Alternatively, we might interpret viability in terms of its being within the basin of attraction for a dissipative, gradient-coupling regime satisfying C1–C4.

⁶ At the same time, attractors can shift or even collapse, which is why we need C5 (selection across generations) to stabilize the attractive landscape on the meta-stability front. Living systems persist, because the attractor geometry, stabilized across generations, makes departure from the basin of attraction a self-correcting exercise.

bounded region such as its own set of viable states $\mathcal{K}$ can be described by a Lyapunov-like, viability-style function over $\mathcal{K}$ (Aubin, Bayen, and Saint-Pierre, 2011; Khalil and Grizzle, 2002; Lyapunov, 1992). Therefore, there has to exist a function $V : S_{\mathcal{L}} \rightarrow \mathbb{R}$ such that V is maximized at states deepest within the viability set $\mathcal{K}$, decreases toward the boundary $\partial\mathcal{K}$, and undefined beyond $\partial\mathcal{K}$. The optimal policy relative to a finite time horizon up to T can be defined in the following manner:

$$\pi_t^* = \arg \max_{\pi} \mathbb{E} \left[\sum_{\tau=t+1}^T V(s_{\tau}) \mid \pi \right]$$

Whereas R determines what can persist in the world, V encodes what must be preferred for living systems such as $\mathcal{L}$ to persist. It is important to note that this V does not yet have value in the psychological sense. It does, however, have the property of inducing a normative distinction relative to the viability conditions C1–C4. Any distinction counts as normative in a minimal sense iff it allows us to distinguish between states (evaluative dimension) and constrains behaviour in the appropriate manner (action-guiding dimension). V has an evaluative dimension, since it allows us to distinguish between better or more viable states (higher V value) and worse or less viable ones (lower V value). V also has an action-guiding dimension, since it is functionally integrated into the life-relevant processes $\mathcal{F}_{\mathcal{L}}$, such that the dynamics of $\mathcal{L}$ is systematically biased toward states with a higher V value. In this way, V does not merely describe viability: it also exerts selection pressure over trajectories in $S_{\mathcal{L}}$.

We are now ready to make sense of valuation in terms of the normative, Lyapunov-like viability function V . Valuation refers to the ordering over states $s \in S_{\mathcal{L}}$ induced by the viability conditions C1–C4 and the viability function V . Formally:

$$s_i \succ s_j \iff V(s_i) > V(s_j)$$

As this ordering over states is independent of both the agent’s model P and her updated beliefs Q and solely determined by the dynamics induced under R , we can infer that R uniquely provides a stable, non-arbitrary, and non-epistemic basis for evaluation. Any adequate grounding of value must ultimately be located in R . At the same time, while valuation is grounded in R , it can still be represented in P and tracked by Q . The generative process R determines the dynamical constraints under which a living system $\mathcal{L}$ persists in accordance with the viability conditions C1–C4, thereby inducing a viability set $\mathcal{K}$ and an associated Lyapunov-like viability function V that orders states in terms of their varying degrees of viability. Active inference, then, provides a schema for policy selection under uncertainty, where expected free energy $G(\pi)$ (§ 3) serves as a useful proxy for maximizing expected viability under V .

In addition, different classes of living systems exhibit a diverse range of strategies and control architectures – sessility versus motility, reflex- versus model-based control, endothermy versus ectothermy, homeostasis versus allostasis, and so on – for maintaining viability under dynamical conditions. Notwithstanding their surface differences, however, these strategies and control architectures possess a common structure: each specifies a distinct way of exerting control over the viability set $\mathcal{K}$ under R . Some systems can alter their viability landscape by internalizing control and modifying their internal dynamics (for instance, by endothermy, homeostasis, and so on). Other systems can navigate within a given viability landscape by externalizing control through action and policy selection (for instance, by ectothermy, motility, and so on). Systems that internalize control tend to realize a viability landscape that is smoother and more robust to perturbations, since small deviations in state or policy do not immediately result in large or dramatic decreases

in viability. By contrast, systems that externalize control tend to rely on a precise coupling to the dynamics of their environment and exhibit sharper gradients in the viability landscape. The persistence of the latter class of systems is sensitive to accuracy in policy selection and the favourability of environmental conditions. Distinct strategies and control architectures (internal versus external, reactive versus predictive, and so on) correspond to different transformations of the viability landscape and there give rise to distinct classes of valuation structures.

On our view, the prior preferences encoded in P do not constitute an independent source of normativity. Rather, they are a reflection or internal representation of how states are ordered relative to the viability constraints ($\mathcal{K}$ and V) induced by R and the system’s control architectures. This allows us to define intelligence as an agent’s capacity to use its generative model P to select optimal policies π for maximizing V . Through her use of intelligence, each agent can better align its actions and control states to the viability structure of the world. Intelligent agents regulate their behaviour in accordance with the constraints imposed by R , coordinating their internal model P with the dynamics that determine persistence. Furthermore, our account of valuation yields a non-arbitrary preference structure grounded in the physical and thermodynamic organization of living systems. This viability-constrained valuation structure is encoded in the generative process R , approximated by the generative model P , and tracked by the approximate posterior Q . Our claim is not that thermodynamics produces value in the psychological sense, but rather that it constrains the space of possible valuation structures available to any living system $\mathcal{L}$ satisfying the viability conditions C1–C4. In particular, given a viability function V , thermodynamic constraints restrict which mappings $V : S_{\mathcal{L}} \rightarrow \mathbb{R}$ can be realized, since they require that higher-valued states correspond to regions of the state

space $S_{\mathcal{L}}$ that can be maintained dynamically under the coupling together of gradients ($\exists \Phi_1 \Phi_2 (C : \Phi_1 \otimes \Phi_2 \rightarrow W_{\mathcal{L}})$).

§7 Conclusion

To conclude, active inference frameworks provide a fascinating account of intelligence and can be represented in terms of the tuple $(P, Q, R, S, A, U, \Omega)$ (§ 2). According to their account of policy selection, the expected free energy function $G(\pi)$, definable in terms of P, Q, R , and Kullback-Leibler divergence D_{KL} , allows an agent to determine whether a particular policy is good or bad (§ 3). At the same time, active inference is confronted with a grounding problem (§ 4):

(Lemma or normative grounding) Any theory of agency that purports to explain goal-directed behaviour must account for the origin of its evaluative structure.

(Theorem concerning the grounding problem) Within active inference frameworks, the generative process R must satisfy exactly one of the following conditions:

(Horn 1 or primitive grounding) R is given exogeneously, with its evaluative structure stipulated;

(Horn 2 or non-primitive grounding) R is derived from the internal mechanisms of active inference frameworks;

(Horn 3 or groundlessness) R is not distinct from the generative model P , and is ultimately reducible to the probabilistic structure continuous with P .

(Corollary or incompatibility result) Active inference frameworks cannot simultaneously satisfy the following conditions:

- Foundational status (violated under Horn 1);
- Absence of circularity or normative regress (violated under Horn 2);
- A genuine distinction between preference and prediction (violated under Horn 3).

We have argued that our theory of intelligence, interpreted as an extension of active inference frameworks, solves this grounding problem (§§ 5–6). In the first instance, this theory is nested within a theory of structure and a theory of life (§ 5). In the second instance, this theory allows us to achieve a non-stipulative grounding of value. Viability, defined relative to R , matters in the context of life, since living things must remain within the viability set $\mathcal{K}$ even in the face of perturbation. The viability function V , encoding what must be preferred for living systems to persist, and the viability conditions (C1–C4) jointly induce an ordering over states $s \in S_{\mathcal{L}}$ that is best described as a valuation. Distinct classes of valuation structures arise as a result of different classes of living systems exhibiting different strategies and control architectures (internal versus external, reactive versus predictive, and so on). Intelligence is the ability that an agent has to anticipate the consequences of her actions under R and align her behaviour accordingly with a viability structure under $\mathcal{K}$ and V . The more intelligent the living system, the greater typically will be her ability to navigate a wider range of environmental conditions, maintain herself under constant perturbation, and realize trajectories that more reliably remain within $\mathcal{K}$.

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